

HATI v2: the sub-resolution hazard heatmap, explained

A walk through every channel that goes into the heatmap, what each one sees, why it sees it, and how they get fused into a single number per pixel that says “this patch is dangerous.” Plus the roadmap, so it’s all in one place.

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What this document is for

If you read the HATI v1.5 paper, you know the system finds lunar hazards down to about 3 m in size. You also know it can’t find anything smaller than that, because the Lunar Reconnaissance Orbiter’s digital elevation model (DEM) resolves features at 1.5 m per pixel. By the Nyquist sampling theorem, anything smaller than roughly two pixels (3 m) can’t exist in the data as a feature a detector can latch onto.

The gap between *what we detect* (≥ 3 m) and *what kills landers* (0.5 to 3 m) is the resolution gap. HATI v2 closes it without inventing data. Instead of pretending to see individual sub-3 m craters, the heatmap produces a per-pixel probability that a patch contains unresolved hazards. The lander doesn’t need to know where every small crater is. It needs to know which patches statistically hold too many of them to land on.

This document explains every ingredient. Intended reader: anyone who’s going to look at this work and ask “how does it actually work.” Collaborators, supervisors, future-me, anyone who needs to sign off on the method.

The problem, in one paragraph

The lunar surface at the sub-3 m scale is dominated by small, degraded impact craters. Cai & Fa (2020) measured this directly: meter-scale roughness on the Moon is the cumulative signature of unresolved cratering. Kreslavsky et al. (2013) found the same thing at slightly larger scales. The signal of these features is already in the 1.5 m/px DEM. Not as individual detectable craters, but as statistical roughness: small variations in slope, in curvature, in how illumination falls on the surface. The signal exists. It just hasn’t been used for landing safety. Planetary scientists used it for geological classification (terrain age, surface type). We’re re-purposing it.

The core idea

Build a stack of “fingerprints,” scalar fields computed from the DEM and from LROC NAC imagery, that each encode a different aspect of unresolved roughness. Combine them into one per-pixel score between 0 and 1 using a weighted sum where every weight is anchored in literature. That score is the heatmap: low means smooth and unlikely to hide hazards, high means rough and statistically dangerous.

The heatmap is generated on the ground, in the Titan segment, alongside the high-confidence hazard map. It feeds into the Mission Safety Layer next to the Ghost Class. Scout’s 1.64 ms inference budget doesn’t change.

Part 1. The DEM-derived channels (Tier 1)

These seven fields come from the 1.5 m/px DEM alone. All of them are computable with NumPy and GDAL. Each one is computed in a sliding window over the tile: for each pixel, look at the local neighbourhood and produce a scalar.

One pre-processing step matters before any of them. Regional slope detrending. A tile sitting on a 5° regional slope has high slope and curvature signatures everywhere, which would saturate every channel. We fit a local planar surface and subtract it. What remains is the roughness on top of the regional geometry.

1.1 Median Differential Slope (MDS) at multiple scales

The MDS measures the curvature contributed specifically by features of size L . Compute the slope at scale L and at scale $2L$; take the difference. What’s left is the curvature signal at scale L . Repeat for $L = 3, 5, 10, 20$ pixels.

A small crater leaves a distinctive curvature signature at its rim. The MDS at $L = 3$ pixels (4.5 m) is sensitive to features right at the Nyquist limit. The MDS at $L = 20$ pixels (30 m) is sensitive to larger features that the standard YOLO detector might already see. Stacking multiple L s gives a roughness spectrum, not a single number.

Anchor: Kreslavsky & Head (2000) introduced this for measuring surface ages. Rosenberg et al. (2011) applied it globally to LOLA data.

1.2 RMS slope at baseline L

Root-mean-square slope in a window of size L . First-derivative roughness. Steeper terrain, higher RMS slope.

A field of sub-3 m boulders or shallow craters inflates RMS slope even when no individual feature is detectable. Rosenberg (2011) showed this scales predictably with surface age and process history.

Anchor: Rosenberg et al. (2011), Cai & Fa (2020).

1.3 Local Hurst exponent (via wavelet leaders)

The Hurst exponent measures the scaling behaviour of roughness. A landscape that looks similarly rough at every scale has a Hurst exponent around 0.5. A landscape with excess fine-scale structure relative to large scale, exactly the signature of unresolved small craters on otherwise smooth terrain, has a lower H.

This is the most direct measurement of “are there features here that I can’t resolve individually but that affect the statistics?” Two patches can have identical RMS slope yet very different Hurst exponents. The one with the lower H is the one hiding the small unresolved hazards.

The obvious way to estimate H, fitting a power law to the FFT-based variogram, is fragile on small windows and noisy. We use wavelet leaders (Lemelin 2020) instead. They give a robust estimate of the scaling exponent even on small patches with non-stationary statistics. This is the one channel where the more complex implementation is worth it, because the simple one is known to misbehave.

Anchor: Rosenburg (2011) for the Hurst exponent; Lemelin (2020) for wavelet leaders.

1.4 Interquartile range (IQR) of profile curvature

How much the curvature varies within a window, measured robustly. The IQR is the spread between the 25th and 75th percentile of curvature values. Robust because it ignores the most extreme outliers.

A crater bowl has a characteristic curvature profile: rim positive, floor negative. A patch with many small craters has high curvature variance. A flat patch has low variance. This catches crater fields even when individual craters are below detection.

Anchor: Kreslavsky et al. (2013).

1.5 IQR of slope

The robust spread of slope values in the window.

Boulder fields and crater fields produce characteristic slope distributions: many small slopes pointing in random directions. A smooth maria has a tight slope distribution (narrow IQR). A boulder-strewn patch has a wide IQR. This channel is particularly sensitive to mixed-feature terrain.

Anchor: Kreslavsky et al. (2013), Wang et al. (2024).

1.6 RMS deviation from planar fit

After subtracting the best-fitting plane in the window, how much residual height remains? Root-mean-square of the residuals.

Features that aren’t aligned with the regional slope, small positive or negative relief, show up as residuals after detrending. This is a direct measurement of “stuff that doesn’t fit the plane,” which is exactly what an unresolved feature looks like.

Anchor: standard geomorphometry; used in Wang et al. (2024).

1.7 Topographic Position Index (TPI), Terrain Ruggedness Index (TRI), curvature suite

A small suite of standard GIS hazard metrics. TPI measures whether a pixel is higher or lower than its neighbourhood (positive = ridge, negative = valley). TRI measures the average elevation change between a pixel and its neighbours. Profile and plan curvature measure how surfaces bend along and across the local slope direction.

These are direction-resolved. A linear ridge looks different from an isotropic boulder field, even if their RMS roughness is identical. The TPI/TRI/curvature suite gives the heatmap geometric specificity.

Anchor: standard in GIS hazard assessment; Wang et al. (2024) used exactly this stack.

Part 2. The LROC NAC channels (multi-illumination)

This is where v2 reaches below the DEM's resolution ceiling. The 1.5 m/px DEM has no information about a 50 cm boulder. But the photons that boulder reflects off the lunar surface end up in LROC Narrow Angle Camera (NAC) imagery, sampled at roughly 0.5 m per pixel. A 50 cm boulder under a low sun angle (6° elevation, typical at the south pole) casts a shadow about 5 m long. That shadow is clearly visible in NAC.

The trick: a single NAC image is just a snapshot. A real shadow and a random dark patch (a noisy pixel, a coincidental low albedo) look identical in one frame. But across many NAC images at different sun angles, a real boulder's shadow moves in a way that tracks the sun. A noise artifact doesn't.

So we collect every NAC image we can find of each tile, at different sun positions, and stack them. Three statistical channels come out.

2.1 Shadow persistence

For each pixel, the fraction of NAC frames in which that pixel is classified as in shadow.

A real topographic feature is shadowed for a predictable subset of sun angles, then sunlit for the others. The fraction is informative about feature height: taller features stay shadowed longer as the sun rotates. A noise pixel has a near-random fraction.

Threshold: within each NAC frame, we use Otsu or local-adaptive thresholding to classify each pixel as shadow or sunlit, then count across the stack.

2.2 Shadow anisotropy

When a pixel is in shadow, is the shadow direction consistent with the sun direction in that frame? For each shadowed pixel, we check whether its shadow extends opposite to the sun azimuth. We compute the agreement across all frames.

A real shadow is cast away from the sun. A noise pixel or a low-albedo region is shadowed in random directions across frames. High anisotropy means a real feature. Low anisotropy means noise or low albedo masquerading as shadow.

2.3 Shadow elongation (relative to sun azimuth)

The length-to-width ratio of each shadow, projected along the sun's azimuth in that frame. A boulder casts a shadow elongated along the anti-sun direction. The elongation is proportional to feature height divided by sun elevation.

This channel directly encodes feature height, which is what makes a feature dangerous. A 10 cm pebble casts a negligible shadow at 6° sun elevation. A 50 cm boulder casts a 5 m elongated shadow that screams.

Part 3. Fusion. Turning ten channels into one number.

Each channel is computed independently and normalized to a regional baseline. The baseline is the Rosenberg et al. (2011) global LOLA distribution of that statistic for the corresponding terrain class (highlands vs. mare vs. polar). After normalization, each channel is expressed as a regional z-score: how many standard deviations is this pixel above or below typical terrain of its kind?

Why regional normalization matters: a south polar tile sitting on naturally rougher terrain than the average maria would otherwise saturate every channel. By comparing against the polar baseline instead of the global one, we measure anomalous roughness against expectation.

Then the fusion equation:

$$H(x) = \sigma(\sum_i w_i \cdot \tilde{f}_i(x) + b)$$

Where:

- $H(x)$ is the heatmap value at pixel x , bounded in $[0, 1]$
- $\tilde{f}_i(x)$ is the i -th channel value at pixel x , regionally normalized
- w_i is the literature-anchored weight for channel i
- b is a bias term
- σ is the logistic sigmoid (gives the bounded $[0, 1]$ output)

The weights are not tuned to maximize any HATI-internal performance metric. They are set from the priors in the literature for each channel's relevance to landing safety. Deliberately. Tuning the weights against Athena outcomes (the planned counterfactual demonstration) would be circular and inadmissible. We want a defensible heatmap before we run any counterfactual.

If, after v2 ships, we want a learned version with better performance, that becomes a follow-up paper. Logistic regression trained on a held-out 70/30 split of the 1,459-object Ghost Class audit, with the literature-anchored weights as prior. Wang et al. (2024) got 96% accuracy with a six-feature XGBoost on a related task, which suggests there is real room above the literature-anchored baseline. But that comes later. The first deliverable is the version whose every coefficient I can defend from first principles.

Part 4. How we know it works (validation)

Three tests, ordered by cost.

V3. Gate test: AUC against the 1,459-object audit

The HATI v1.5 forensic audit confirmed that 89.5% of “false positive” detections from Titan were physically real hazards (vertical relief > 15 cm). We record the heatmap value at the centroid of each of the 1,459 audited objects, plus 1, 3, and 10 pixel offsets. We compute the area under the ROC curve for using the heatmap value as a discriminator between the 1,306 valid hazards and the 153 flat-ground noise objects.

If $AUC > 0.7$, the heatmap correlates with real physical hazards. If $AUC < 0.6$, the fusion is wrong and we re-tune the weights. This is the gate. One number.

V1. Primary test: Ripley's K with conditioned null

Treat the Ghost Class detections as a marked point process. Run a Ripley's K spatial correlation test: do Ghost Class detections cluster inside upper-quartile heatmap pixels at the right spatial scales? We generate 99 Monte Carlo randomizations of detection positions (conditional on Titan's overall detection field) and compute envelope bounds. The test passes if the observed $K(r \mid H > \tau)$ exceeds the unconditioned $K(r)$ significantly for $r < 30$ m.

To my knowledge no prior lunar landing safety work has used Ripley's K validation. The conditioning on Titan's detection field is critical. Without it we'd partly be measuring Titan's own bias rather than the heatmap's predictive power.

V2. Optional: hand-counted NAC patch comparison

On 5 to 10 small LROC NAC patches at 0.5 m/px, hand-count the visible sub-3 m craters and boulders. Then check whether the heatmap value in each patch correlates with the hand-counted feature density. Target Spearman $\rho > 0.5$ with DEM channels alone, > 0.6 with NAC shadow channels added.

This is the most direct ground-truth test but requires manual geological work. Nice-to-have rather than blocking. If V1 and V3 both pass, V2 strengthens the paper but isn't required.

Part 5. How this plugs into HATI

The heatmap is additive to the existing v1.5 architecture. It doesn't change the asymmetric split-segment design, the Scout latency budget, or the cost function in §6 of the v1.5 paper.

What changes:

Titan's output expands. Instead of just the hazard map (`titan_hazard_map.geojson`), Titan now also writes a hazard heatmap (`titan_hazard_heatmap.tiff`) aligned to the DEM frame in standard GeoTIFF. This is consumed by the same Mission Safety Layer the Ghost Class feeds into.

The guidance law gets one more input. The random-walk divert logic (§5.2 of v1.5) currently treats every pixel outside an explicit hazard box as equally safe. With the heatmap, it weights the divert search by $1 - H(x)$. The lander still searches for “negative space between hazard clusters,” but now the negative space is statistically scored.

Scout’s onboard work is unchanged. No new model, no new inference, no new latency. The heatmap is purely a Titan-side product that arrives onboard pre-loaded along with the hazard map.

The asymmetric cost function carries forward. False positives (avoiding a low-heatmap region unnecessarily) cost negligible fuel. False negatives (landing in a high-heatmap region with unresolved hazards) cost the mission. The heatmap is tuned philosophically to be conservative under this asymmetry: when in doubt, flag.

Part 6. What this is not

Honest framing. The heatmap:

Doesn’t solve navigation. If the lander doesn’t know where it is, no map helps. Athena and Odysseus both had navigation-driven failure modes that no perception layer could have prevented. The heatmap is a perception prior, not a navigation system. The terrain-relative navigation work proposed for v2 is a separate contribution that addresses navigation; the heatmap addresses perception.

Doesn’t detect individual sub-3 m hazards. It produces a probability that a patch contains them, not a list of boxes around them. The guidance system uses the probability; it doesn’t expect specific feature locations.

Doesn’t replace the v1.5 hazard map. Where Titan already detects features at ≥ 3 m, those detections remain authoritative. The heatmap is a complement, filling the gap below detection.

Isn’t magic super-resolution. Unlike the GAN approaches we reject in v1.5 §2.3, the heatmap never invents a feature that isn’t in the data. It reports statistical evidence for unresolved hazards based on physical roughness signatures that decades of planetary science have documented in the DEM and in multi-illumination NAC imagery. Every output is a function of measured photons or measured heights, not generated content.

Isn’t the final word. A v3 paper using XGBoost or a small CNN with the literature-anchored heatmap as initialization will almost certainly beat the literature-anchored version on raw AUC. That is fine. The point of v2 is to ship a method whose every coefficient is defensible from first principles, validated against an independent forensic audit, and decomposable per pixel.

Part 7. Roadmap

This is the sanity check. Where we are, what’s next, what comes after.

Where we are

HATI v1.5 is complete. The manuscript is locally archived and waiting for Iris Fernandes's final sign-off before publication. The code is on GitHub at [Cyrex567/HATI](#). The Athena counterfactual hasn't been run yet because the tool that should run it (the v2 heatmap) doesn't exist yet.

What's next: build v2

Two contributions land together in v2.

The first is the sub-resolution hazard heatmap described in this document. Ten channels: seven DEM-derived, three from LROC NAC multi-illumination shadow statistics. Literature-anchored fusion. Validated against V1 (Ripley's K) and V3 (AUC on the 1,459-object audit) before any counterfactual runs.

The second is terrain-relative navigation (TRN) using the shared detection vocabulary between Titan and Scout. Both speak the same language (same YOLOv8-OBb architecture, same crater and boulder classes). That means Scout's live detections can be matched directly against Titan's landmark database for position estimation. The heatmap participates as a prior on expected detection density, letting the matcher reject false correspondences in regions where Scout is expected to under-detect. This addresses the failure mode (navigation) that the heatmap alone can't.

The validation milestone: the Athena counterfactual

Once both contributions are in, run the 100-mission LM-7 Monte Carlo on the actual Intuitive Machines IM-2 (Athena) landing site at Mons Mouton. With and without v2 active. Report a prevention rate with a Wilson 95% confidence interval.

If the prevention rate is meaningfully above zero, we have a published, defensible answer to "would HATI have helped Athena?" If it's not, we have an honest negative result and we know where the architecture breaks down.

What success looks like

V3 AUC ≥ 0.7 . V1 Ripley's K passes (clustering significant at $r < 30$ m). TRN matcher converges on at least 90% of synthetic descent frames in the validation set. Athena counterfactual reports prevention $\geq 50\%$ with confidence intervals that don't cross zero.

If we hit those numbers, v2 is publishable as a method paper with the Athena counterfactual as the application. If we miss one, we know exactly which subsystem is the bottleneck.

Realistic timeline

Tier-1 DEM heatmap pipeline: 5 working days. NAC shadow stat pipeline: 5 working days. Fusion + per-region normalization: 1 day. V3 + V1 validation: 3 days. TRN matcher (RANSAC correspondence + state estimator): 7 to 10 days. Athena counterfactual run + analysis: 2 days. Paper rewrite: 5 to 7 days.

Roughly 12 weeks of focused work alongside coursework. Realistic calendar time: 4 to 5 months if I can give 25 hours a week, longer if not.

What's deliberately not in v2

The Tier-2 channels beyond NAC shadows (Diviner thermal IR rock abundance, Hapke macroscopic roughness, PSD allometric breakover) are not in scope. They would improve performance but balloon the methodological surface area. Same with the XGBoost or CNN upgrade to the fusion model: deferred. The point of v2 is one solid contribution with literature-anchored coefficients, not a kitchen-sink result.

If v2 lands clean, those become v3.

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Companion document to the HATI v1.5 manuscript and v2 technical memo. For questions or corrections, talk to me.